

SIEMENS ENERGY EHS PLAN

Client : MoE & BGC

Project : KAZ Power Plant Upgrade

Date : 03.03.2026

Revision: 00

Author : Dogukan Arandi

Accepted by : Onur Ozvatan

Ozvatan
Onur

Digitally signed by Ozvatan Onur
DN: cn=Ozvatan Onur, c=DE,
o=Siemens,
email=onur.ozvatan@siemens-
energy.com
Date: 2026.03.04 11:10:18 +03'00'

Arandi
Dogukan

Dijital olarak imzalayan Arandi
Dogukan
DN: cn=Arandi Dogukan,
o=Siemens,
email=dogukan.arandi@siemens-
energy.com
Tarih 2026.03.04 11:02:32
+03'00'

Table of Contents

1	Purpose	3
1.1	Scope and understanding of EHS – Environment, Health and Safety	3
2	Project Description	4
3	Our Management System	5
4	Administrative Matters	5
4.1	The project's EHS organization and personnel	5
4.1.1	The Construction Client	5
4.1.2	The Principal Undertaking	5
4.1.3	Sub-Contractors	6
4.2	Project Organization	6
4.2.1	The Project Manager	6
4.2.2	The EHS Manager	6
4.2.3	The EHS Representative	6
4.2.4	The Site Manager	7
5	Objectives for the Project	8
5.1	The project's EHS policy and objectives	8
5.2	Target	8
6	Planning	8
6.1	Risk Management	9
6.2	Risk Management during the Construction Period	9
6.3	Before start of Working	10
7	Site Conditions	11
7.1	Communication	11
7.1.1	Kick-off Meeting	11
7.1.2	Morning meeting/ Toolbox-meeting	11
7.1.3	Safety Walks	11
7.1.4	Reporting	11
7.2	Working Hours	11
7.3	Work Equipment	12
7.4	Safety Work	12
7.5	Working in Heights	12
7.6	Hot Work	13
7.7	Injuries, RUO and Improvements	13
7.8	Violation of the Regulations	13
8	External Environment	13
8.1	Environmental, Transport and Site Plan	13
8.2	Chemicals and Substances are harmful to Health	13
8.3	Tidiness and Waste Management	14
9	Electrical Safety	14
9.1	Safety at Worksite	14
9.2	Activities near Power Lines and Cables	15
9.3	Work during Thunderstorms	15
10	Foreign affairs	15
11	Attachments	15

1 Purpose

The purpose of the EHS Plan prepared for the **KAZ Power Plant Upgrade** project is to define the Occupational Health and Safety (EHS) and Environmental management methods used in the project.

Safety for people is an absolute goal for our business, at the same time as we want to help improve the external environment - Our projects must be carried out without harm to life, property, and the environment.

The aim is to provide preventive measures for and primarily to avoid, secondarily to limit the consequences of undesirable incidents/ accidents, so that safety for life and health, the external environment and property are safeguarded as well as possible.

We review this EHS plan with all employees and subcontractors before they arrive at site.

1.1 Scope and understanding of EHS – Environment, Health and Safety

Sustainable Solutions:

Sustainability is at the core of our company strategy, enabling us to drive future growth, improve prosperity, and increase our resilience. We are a partner of choice for our stakeholders and shaper of the energy transition worldwide.

Our Sustainability Program is oriented toward our company's purpose, "We energize society" whilst contributing to the United Nations' Sustainable Development Goals. Through it, we guide our activities and focus on the most relevant topics that help us achieve our mission to support our customers in transitioning to a more sustainable world.

Our businesses, regional entities and central functions work together to implement the program. We measure our progress against our goals and strategic KPIs.

As a global actor, we focus on the following areas in the UN's Agenda 2030:



Siemens Energy aims to be carbon neutral in 2030. How we meet the UN Agenda 2030, safeguard sustainability and do our best to combat climate change is described in our [Sustainability Report](#).

Health: Comprises physical or mental health, including injury and illness as a result from work. All project participants shall have access to medical treatment and particularly vulnerable groups must be identified and followed up. Health includes the working environment with respect to contributions/ co-determination, relationships with colleagues and content in work tasks.

For the construction project, this means working actively to ensure that all project participants experience a working environment that allows him/her to leave the workplace safe and unharmed, as well as feeling satisfied and well at the end of the working day. The worker shall have influence on work processes and decisions, where there is room for both praise and criticism, and the project shall have a working environment characterized by openness and loyalty among the participants.

Environment:

Comprises the environmental impact throughout the project's entire life cycle, from the choice of technology/ products, methods of production/ construction, emissions during the operating period and finally phasing out/ recycling.

For the construction project, this means that active efforts will take place to reduce the climate footprint and pollution of the construction site, surrounding areas, roads and the neighborhood. We will recommend the most sustainable solutions where the project has a choice, be it products, equipment, construction method or schedule. Pollution might follow from bad choice of products, logistics or construction methods and lead to unnecessary light, noise, vibrations or emissions, with associated damage to communities, flora and fauna, locally or globally.

Safety:

Comprises safety for workers and visitors to the site, passers-by and neighbors.

For the construction project, this means that active efforts will take place to gain control over hazards that may threaten people, the environment and surroundings/ property. Safety requires good interdisciplinary coordination, good plans and a good safety culture, and is reflected in the priorities that the project must constantly undertake.

2 Project Description

KAZ Power Plant Upgrade Project represents a comprehensive modernization and expansion of the existing 400/132/11 kV Air Insulated Switchgear (AIS) facilities located in Khor Al-Zubair. The project scope encompasses all civil, mechanical, and electrical works required to install new 400 kV and 132 kV bays, including primary equipment such as circuit breakers, disconnectors, earthing switches, current transformers, voltage transformers, surge arresters, and associated busbar extensions. Additional scope elements include the construction of new foundations, enhancement of the earthing grid, installation of steel structures, and execution of all related cable works covering control, protection, metering, telecom, and SCADA/DAS interfacing systems.

The upgrade is strategically vital for improving grid reliability, supporting regional power demand growth, and enhancing transmission stability across the Basra region. Execution of the works requires coordinated planning with ongoing Ministry of Electricity (MoE) operations to ensure uninterrupted power supply and safe working conditions within a high-voltage environment. Key activities involve excavation, heavy lifting, hot work, working at height, and phased energization, which demand rigorous EHS controls and structured risk management.

The project aims to deliver a fully integrated and safely commissioned switchyard extension that meets all technical, contractual, and regulatory requirements. Emphasis is placed on Zero Harm principles, adherence to IEC standards, and maintaining seamless coordination between Siemens Energy, MoE, and BGC throughout construction and commissioning phases.

3 Our Management System

Siemens Energy is a multinational listed company and is subject to laws in Finland and Europe. Siemens Energy Oy is certified in accordance with ISO 9001, ISO 14001 and ISO 45001. The company's IT systems are certified in accordance with ISO 27001.

The management of the project is carried out in accordance with our IMS - Integrated Management System, which follows Siemens Energy's global standard for project implementation.

Environmental challenges are global, and our goal is for all economic growth to come as a result of a high standard of environmental protection and social responsibility. Sustainable development concerns us all, and the company's overall goal is to contribute to sustainable development.

EHS and Sustainability are integrated into all our business processes, and compliance and implementation of all applicable statutory requirements within environmental protection, safety, health and working environment is mandatory.

Read more about Siemens Energy here: www.siemens-energy.com.

4 Administrative Matters

4.1 The project's EHS organization and personnel

Below is a list of the main functions in the project's EHS organization and their area of responsibility. The project's specific organization chart shows the main structure and is included in our quality plan.

4.1.1 The Construction Client

The client's plan for Occupational Health and Safety (EHS plan) is prevailing in the project. We follow the plan and the instructions given by the Construction Client or the Coordinator.

Our work will be planned and conducted according to the EHS-plan, which will be distributed to our project participants before any work begins. We will inform the Construction Client if we detect any deviation that may influence employees' health and safety or the environment.

4.1.2 The Principal Undertaking

Where several companies have simultaneous activity at site, the individual companies' health, environmental and safety work must be coordinated across the companies in order to ensure the safety of the employees. The Principal Undertaking (PU) shall be responsible for this coordination.

Furthermore, the operational responsibility for any workplace must be determined. The operational responsibility includes i.e. crisis management, area security/ access control and waste management. On construction sites, the Construction Client normally contracts this with PU, but in facilities during operation it may also be the Construction Client himself. If the operational responsibility is assigned to a contractor, this is done in a separate agreement/ contract.

In transformer stations, the operational responsibility for the area after the high-voltage system has been adapted for voltage must be clearly stated in the plans. Normally this belongs to the Client's Operations Manager.

4.1.3 Sub-Contractors

All subcontractors shall use their own system for EHS, enriched with any project specific requirements from the Construction Client and us.

All deviating conditions shall be reported. Deviations and RUO's (Report on Unwanted Occurrences) that are reported shall be announced to us on an ongoing basis. A document has been prepared.

4.2 Project Organization

The project's specific organization map shows the main structure.

The usual safety functions for the construction industry and according to Occupational Safety Act are taken care of in our subcontractor's standard function descriptions in the companies' respective quality systems. Relevant safety functions are included in the project organization.

4.2.1 The Project Manager

The Project Manager (PM) has overall responsibility for the project's EHS and quality work, this includes EHS and sustainability.

4.2.2 The EHS Manager

This is the EHS (Health, Safety, Environment) officer appointed by the Department Management to which the project is affiliated.

- During the project proposal phase, they review the EHS requirements and demands specified in the customer specifications.
They support the project proposal manager by analyzing the EHS risks and costs of the project.
- As a partner, they support the project manager by planning, supervising, monitoring, and reporting on the prevention of EHS-related hazards and risks related to the project. In this context, they perform their duties in accordance with the information and reports from the Project EHS Manager/Occupational Safety Specialist and in line with the standards and Safe Working Method determined by Siemens Energy Global.
- They conduct the EHS kick-off meeting organized by the Project Manager.
- They prepare and distribute the project's EHS plan with the project management.
- They assign the EHS Officer who will work on the project to the Project Manager together with the Project Manager.
- They work in accordance with the provisions of the Internal Occupational Health and Safety Directive.

4.2.3 The EHS Representative

The EHS representative is the EHS manager's representative at the construction site. The EHS representative has construction expertise and must ensure daily that the EHS activities at the construction site have the prescribed standard in the plans. He/ she is authorized to correct and stop work.

In our contract, the EHS representative shall, among other things:

- Status report to EHS manager, site manager or project manager
- Be our local contact with the Client's EHS/ OHS organization
- Make sure that all certificates/ approvals are in place for personnel, machines and equipment
- Ensure that the pay and working conditions of the company's subcontractors are in accordance with the applicable general regulations
- Ensure that EHS deviations are reported and followed up in the project's management systems
- Participate in risk assessments and SJA
- If necessary, carry out safety walk in own contract area

Where appropriate, this role can be taken care of by the site manager.

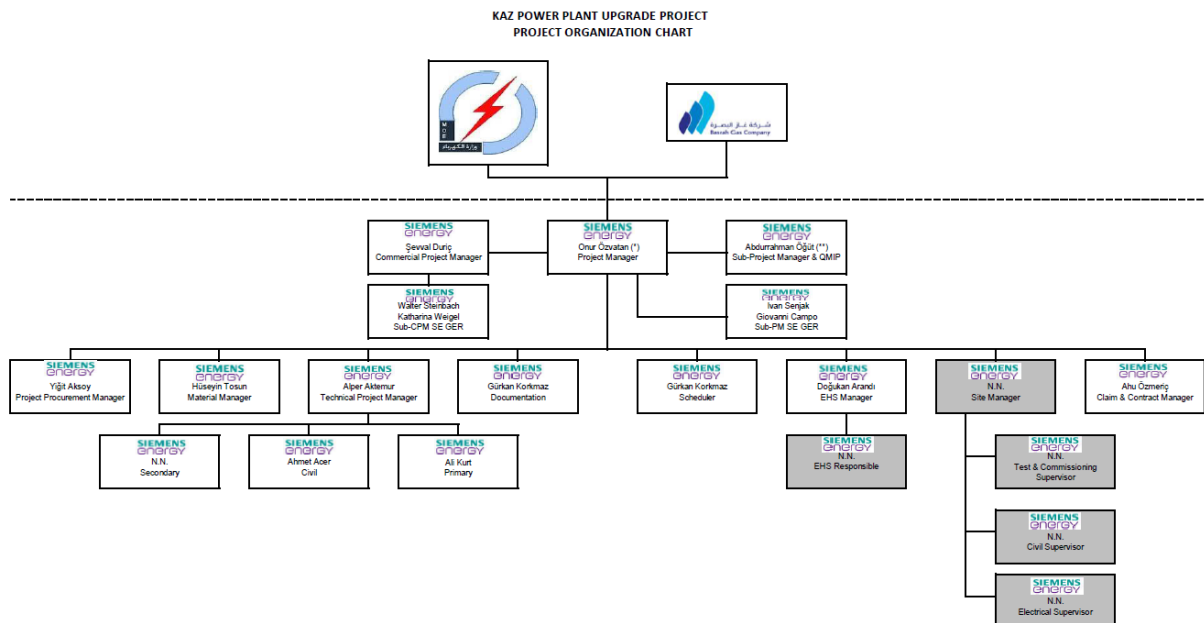
4.2.4 The Site Manager

The Site Manager (SM) is the project manager's representative at site and carries out the day-to-day management of the activities at site. He is responsible for ensuring that the site's EHS, quality and progress follow the applicable standards and plans and is responsible for the daily contact with the Construction Client's representatives and other stakeholders at the site.

The Site Manager's main task within EHS is to ensure that the project's EHS plan is implemented and complied with in all behavior and all activities at the site. In our contract, the Site Manager must ensure this in, among other things:

- work planning
- risk assessments
- safety rounds
- reporting

The Site Manager shall stop any work that exceeds acceptable risk.



5 Objectives for the Project

5.1 The project's EHS policy and objectives

Our standard for EHS should be excellent and exceed customer expectations. The project will:

- promote a Zero-Harm Culture that prevents damage to the environment, people and equipment
- contribute to taking care of project participants with emphasis on preventive measures and training in occupational health and safety (OHS)
- have an open and constructive communication with all stakeholders, based on mutual trust and understanding
- design solutions so that the construction work can be planned, method described and carried out with good rig conditions, simple material handling and safe access in the various phases of the project execution
- do not put unnecessary strain on the neighbors
- Interventions that must be made in nature shall be minimized and nature returned as far as possible to the origin
- choose sustainable construction materials and technical solutions
- deliver a final product that can be operated in a safe and sustainable manner

In order to achieve these goals, it is important to work closely with both the Construction Client and other contractors at the construction site. This collaboration is also crucial for achieving loyalty and commitment from all project staff. The project's priorities must therefore be:

- consideration of the environment
- focus on cooperation with all actors in the project
- proactive EHS work
- keep the workplace clean and tidy – Always!
- efficient solutions for rig and operation
- information and communication

5.2 Target

The project will plan and implement necessary measures to ensure zero harm during the project.

The project's goal for the zero H¹ = ZERO.

6 Planning

The EHS work is planned and organized according to government regulations, and requirements from the construction client and us. EHS activities shall be included in the project's general plans, which include safety rounds, risk assessments, planned Safe Job Analyzes and EHS audits.

EHS work must be proactive. Preventive work must be done by seeking to avoid damage to the environment and the occurrence of dangerous conditions and near miss. Where EHS, progress and finances seem in conflict with each other, EHS shall prevail.

The qualification requirements and safety responsibilities of people who participate in the work or who may be affected by the work are determined in accordance with national legislation.

When performing electrical work, good installation practice and the currently valid laws, regulations, electrical equipment and installation standards and the Client's instructions are followed. All people participating in the work must have a valid occupational safety card (occupational safety in a shared workplace).

6.1 Risk Management

Risk management is established according to principles and guidelines in ISO 45001 and forms a central and proactive management process to reduce the project's negative risk and strengthen the project's opportunities.

The aim of the risk assessment is to identify and evaluate the risk associated with the work and provide a basis for determining measures that must be implemented so that employees can carry out their work in a safe manner. Risk = Probability x Consequence

We always perform risk assessments:

- during planning
- with our subcontractor before his activity starts on the construction site
- with our own test engineers before commissioning

The risk picture is dynamic and adapts to the progress on the construction site, the risk assessment must therefore be updated at all planning meetings after the start of construction. However, it is difficult to plan in detail particularly long in advance and Safe Job Analysis (SJA) is therefore always a relevant tool for occurring work tasks that are defined as dangerous by the project management or the person responsible for planning and/or performing the task (The Last Planner).

6.2 Risk Management during the Construction Period

Morning meetings/ Toolbox meetings (See section 7.1.2) form an important part of the work to reduce general risk. Meetings shall be held at the beginning of the working day with relevant topics for the forthcoming work. Content and participation must be documented.

For work close to live high-voltage systems or when our delivery is ready to come under voltage, our work will normally be subject to the safety regulations from the Client's local Operations Manager. Nevertheless, we have independent responsibility for our employees and subcontractors, and our own project management must therefore ensure the quality of the safety measures available from the Operations Manager before any such work is performed.

Everyone who works on our projects has a duty to stop dangerous work. In such a case, the client will be immediately made aware of this. The person who stops dangerous work shall in no way be prosecuted, punished or harassed for the consequences this entails.

6.3 Before start of Working

Competence/ training/ certificates

When the system is - or may come - under voltage, the procedure on Electrical Safety apply. The regulations require that everyone who is given unaccompanied access to the actual areas has valid safety- and first-aid training.

Before starting up at the construction site, all personnel in our contract must be given training in the following:

- PSI – Personal Safety Instruction
- Our EHS plan
- Information about the Client's EHS-plan
- Electrical safety
- General EHS rules and requirements for PPE (Personal Protective Equipment)
- Special project requirements
- Emergency plans and routines
- Any additional requirements from the Construction Client

The safety training shall be documented.

Documentation of tools and equipment

Before starting up the construction site, we will together with our subcontractors establish a system that ensures that relevant documentation is available on the construction site. The following documentation applies:

- Completed project-specific safety training
- Relevant equipment is approved/ marked/ certified/ calibrated
- Annual/ periodic inspections of equipment, tools and machines
- Relevant certificates/ training for the use of equipment, tools and machines
- Relevant certificates/ training for special work, as required by law/ regulations

This documentation shall be available in due time before performance of the actual work.

Personal protective equipment

The project staff shall use personal protective equipment (PPE) in line with our requirements and any additional requirements from the Construction Client. As a minimum, we require the following PPE:

- Helmet with chin strap or protective cap
- Hearing protection
- Flame retardant protective clothing (Trousers and jacket in visibility class 1)
- Safety shoes with toe protection and nail tread (S3/S1p)
- Safety glasses
- Protective gloves

In addition, other PPE may be required for special types of work operations, according to SJA, project-specific procedures and/or regulations.

7 Site Conditions

7.1 Communication

The project's EHS should be the first topic at all formal meetings between the parties involved at the facility or in the office. This shall be stated in the report from the meeting.

The language on the construction site is Finnish. All employees must either speak a common language or work with a companion who speaks both a Scandinavian language and the relevant foreign language.

Only the Construction Client shall comment to the media and public and private stakeholders regarding the project. We will request permission on behalf of participants in our contract, if applicable.

7.1.1 Kick-off Meeting

Before starting up at the plant, a kick-off meeting will be held to ensure that the project's safety approach is complied with at the plant.

- The need for coordination with side-contractors shall be clarified
- The EHS regulations and emergency procedures are included

Content and participation must be documented.

7.1.2 Morning meeting/ Toolbox-meeting

Daily meetings with relevant topics for the upcoming work at the beginning of the working day. Content and participation must be documented.

7.1.3 Safety Walks

Safety walks will be coordinated by the Principal Undertaking and all contractors at the facility are obliged to participate with a representative, and to comply with the orders given hereof.

7.1.4 Reporting

The main contractor is responsible for the site-specific weekly inspections and documents the inspections on MVRS forms in accordance with the client's instructions (HeadPower IWM digital reports or Enablon).

7.2 Working Hours

Access to the site must be agreed and planned with the Construction Client, who must ensure safe access and make the necessary security functions available to the project, corresponding to the project's needs and working hours. Work is planned at the site at 08-17.00, 6 days a week.

The sub-contractor's working hours scheme will be approved by Siemens Energy.

7.3 Work Equipment

The use of work equipment is regulated in the 'Regulations on the performance of work', and our project plans are in accordance with this. In our contract, all fitters have basic competence which includes mastering standard work equipment that may involve a risk for the user, any additional training to this is documented.

7.4 Safety Work

Safety walks, frequency and scope (areas, participation) are planned and coordinated by the Principal Undertaking. Different periods in the project may have different needs and frequencies.

Our own safety work will in particular focus on:

- Working hours and rest time
- Use of PPE
- Sub-contractors, non-electrical professionals
- Work on/ near live systems
- Tidiness at the workplace
- Work in heights
- Use of SJA - Safe Job Analyzes
- Use of RUO - Report of unsafe occurrences

7.5 Working in Heights

Fall from heights is the most frequent cause of personal injury in working life. Workers working at heights shall be protected against fall accidents. Work at heights means working in scaffolding, ladder, curtain stairs, lift, crane, basket etc.

- When the working place is above the ground, either an approved scaffolding/ work platform or an approved fall protection device must be used. Fall protection in the form of harnesses shall be used when there is no other satisfactory protection against falls.
- Safety harnesses must be used during the installation of railings and during work railings not mounted.
- Scaffolding must be marked in accordance with 'Regulations on the performance of work'. Ladders, stair ladders, curtain stairs, etc. must be clearly marked with the owner's company name and type approval
- Workers must be protected against falling objects from above:
 - Materials and equipment necessary to perform work at height must be adequately secured to prevent weather and wind from setting it in motion.
 - The area below/ near work at height must be cordoned off to avoid work at several levels, possibly workplaces and entrances must be secured.
 - However, work that needs to be carried out under workers at height must be protected against objects falling from above.
- Equipment that is defective and/ or without the required marking must be removed from the site.

Scaffolding must be mounted, used and checked in accordance with requirements in 'Regulations on the performance of work safety act.

7.6 Hot Work

An employee who performs work defined as hot work must have a valid certificate for such work. When carrying out hot work operations, extinguishing equipment shall always be present.

7.7 Injuries, Research Use Only and Improvements

Our project management is responsible for establishing and operating a safe workplace. All unwanted occurrences (approach, condition and event) must be reported. Reporting and processing shall take place without unnecessary delays.

- We use Siemens Energy's portal for reporting (Enablon) non-conformities, incidents and improvement proposals.
- Collaborators and subcontractors use their own systems and report on to our project management

In the event of an emergency or a serious incident with damage to personnel, equipment, or the environment, as well as breaches of laws and regulations, the project must immediately notify in accordance with the project's notification plan. The site manager is responsible for ensuring that the notification plan is known to everyone who performs work in our contract at the site.

We conduct a separate review of serious incidents in accordance with Siemens Energy's procedures for reporting and investigating serious incidents.

Our subcontractors report their incidents accordingly.

7.8 Violation of the Regulations

In the event of a breach of EHS requirements and orders, we can demand that the person in question (sub-contractor, employee) be expelled from the facility.

An employee who encounters intoxication, fight and theft at the workplace shall be expelled immediately, and a personnel case initiated.

8 External Environment

8.1 Environmental, Transport and Site Plan

The Construction Client shall prepare and implement the Environmental, Transport and Site Plan for the site.

We are obliged through the awarded contract to comply with the requirements arising from this.

8.2 Chemicals and Substances are harmful to Health

Requirements for work with chemical and biological risk factors are regulated by the Regulations on the performance of work.

We use the electronic substance register

- Partners and subcontractors use their own systems
- Safety data sheets for relevant substances must be available where the substance is in use

Some very dangerous substances, such as asbestos and polychlorinated biphenyls (PCBs), are banned. The Client shall have an asbestos survey carried out at the work site (the network operator's equipment and premises) if necessary. Existing deposits in old building parts can still occur, and such deposits are subject to strict restrictions and must be reported and treated by certified personnel.

8.3 Tidiness and Waste Management

Safe handling of construction and civil engineering waste is essential for safeguarding both the external environment and the site's safety, health and working environment. Preventive behavior is always having a tidy workplace; Clean everything - always!

General rules for order and tidiness:

- The staff room must be kept tidy
- The area for storage of various materials must be marked and adapted for such use
- Materials, tools and equipment must be stored so that they do not pose a danger to third parties, wildlife or the environment.
- Excess material and packaging must be removed as soon as possible to prevent waste from blowing into the surrounding landscape.
- Roads and transport areas must be kept open and free of material and waste

In other respects, we comply with the site's rules for clearing and waste management, as determined by the Construction Client/ Principal Undertaking.

9 Electrical Safety

9.1 Safety at Worksite

When the system is - or may come - under voltage, the procedure on Electrical Safety apply. From then on, all work taking place in the exposed construction areas shall be carried out in accordance with the current regulations, and anyone given unaccompanied access to the relevant areas shall have the training required by the regulations, Siemens Energy and the Operations Manager.

For the high-voltage system, the exact time "When the system is - or may come - under voltage" determines who is responsible for the operation of the system. Normally, the operational responsibility then falls on the Client's Operations Manager.

For every electrical installation, a high voltage or low voltage Operations Manager (nominated person in control of an electrical installation) must be nominated to be responsible for the operation and maintenance of the installation. Administrative measures must be taken to ensure that the distribution of responsibility does not come into conflict with work on the electrical installation or with its operation.

A high voltage/ low voltage Safety Supervisor must be nominated for each work task. In high voltage installations, a Switching Supervisor must also be nominated if any switching tasks are required.

9.2 Activities near Power Lines and Cables

If work is carried out closer than 30 m from a high-voltage system, the contractor must contact the plant owner in good time to agree on the necessary safety measures, so that he and the plant owner can follow up that electrical safety is taken care of for both parties. In addition, the work area must be physically blocked off or marked.

9.3 Work during Thunderstorms

All work that can cause danger due to lightning and induced voltages (current passage), i.e. installation work, work with a crane, work with a lift, etc. must be interrupted in thunderstorms.

10 Foreign affairs

As a global player and supplier of high technology, we often use foreign specialists from our own company in parts of the execution at the site.

All personnel under our contract must have a written employment agreement comprising pay and working conditions for work performed under the contract, according to ILO Convention 94 on employment clauses in public employment contracts

- Foreign subcontractors and/ or own personnel working in Finland must have a valid work and residence permit
- Electrical professionals must have valid training according to 'the procedure on Electrical Safety
- All equipment used must be approved in accordance with SE procedure

11 Attachments

1. Risk Assessment (Published as a separate document)
2. Instruction_EHS General Requirements for Contractors_EN
3. Standard_EHS Principles and core responsibilities_EN